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Deep Learning-Based Detection of Water Leaks in Urban Distribution Networks Using a CNN-BiLSTM Architecture with Self-Attention

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ABSTRACT This paper presents an end-to-end deep learning framework for automatic leak detection in urban water distribution networks using multivariate sensor time-series data. The proposed approach integrates domain-informed feature engineering, sliding-window sequence modeling, and a hybrid architecture combining Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM), and a self-attention mechanism.

The CNN component captures local temporal patterns, the BiLSTM models long-range dependencies, and the attention mechanism highlights the most informative time steps within each sequence. In addition, a leakage-free temporal data splitting strategy and threshold calibration are employed to ensure robust and realistic evaluation under class imbalance. Experimental results demonstrate that the proposed method achieves strong performance, with an AUC-PR of 0.9583 and an F1-score of 0.9318, while maintaining high recall. Furthermore, benchmark comparisons against classical machine learning models and alternative deep learning architectures confirm the effectiveness of the proposed hybrid model.

Overall, the proposed framework provides a practical and scalable solution for real-world smart water monitoring systems, contributing to improved infrastructure reliability and reduced water loss.

KEY-WORDS Water Leak Detection, Deep Learning, CNN-BiLSTM, Self-Attention, Multivariate Time Series, Sliding Window, Imbalanced Classification, Threshold Calibration, Benchmark Comparison, Smart Water Networks;

I. INTRODUCTION

WATER distribution networks are critical infrastructures for urban environments, ensuring the continuous delivery of potable water to residential, industrial, and commercial users. However, these systems are highly susceptible to leaks due to aging infrastructure, pressure fluctuations, and environmental factors. Undetected leaks can lead to significant water losses, increased operational costs, reduced service efficiency, and long-term structural damage. Consequently, the development of reliable and automated leak detection systems has become an important research and industrial challenge.

Traditional leak detection methods typically rely on manual inspection, acoustic sensing, or rule-based thresholding techniques applied to pressure and flow measurements.

While these approaches can be effective in controlled scenarios, they often lack scalability and adaptability in complex urban networks. Moreover, such methods struggle to capture the temporal dynamics and nonlinear patterns associated with leak events, particularly in the presence of noise and varying operating conditions.

With the increasing deployment of sensor networks and Internet of Things (IoT) technologies in smart water systems, large volumes of multivariate time-series data are now available. This has opened the door for data-driven approaches, particularly deep learning methods, which have demonstrated strong capabilities in modeling sequential data. Architectures such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks have been widely used for time-series classification tasks due to their ability to capture local and long-range dependencies, re-

spectively. However, existing approaches often rely on single-model architectures and lack comprehensive evaluation against alternative methods, limiting their ability to fully exploit complementary modeling strengths.

In this work, we propose a hybrid deep learning architecture that combines CNN, Bidirectional LSTM (BiLSTM), and a self-attention mechanism for automatic water leak detection. The CNN component extracts local temporal features from sensor signals, while the BiLSTM captures bidirectional dependencies across time. The integration of a self-attention layer allows the model to focus on the most informative segments of the input sequence, improving both detection performance and interpretability.

Beyond the model architecture, we design a complete end-to-end pipeline tailored for real-world deployment. This includes domain-informed feature engineering, sliding-window sequence construction, and a strict temporal separation between training, calibration, and test sets to prevent data leakage. Furthermore, we address the inherent class imbalance in leak detection by adopting evaluation metrics such as AUC-PR and by performing threshold calibration to control the trade-off between detection sensitivity and false alarm rate.

To ensure a rigorous evaluation, the proposed model is compared against several baseline machine learning and deep learning approaches. This benchmark analysis provides a comprehensive assessment of the effectiveness of the hybrid architecture relative to alternative methods.

Experimental results on a real-world dataset demonstrate that the proposed approach achieves strong performance, with high AUC-ROC and AUC-PR scores, as well as a favorable balance between precision and recall. In particular, the model maintains a low false alarm rate while achieving high leak detection accuracy, making it suitable for practical monitoring systems where both reliability and responsiveness are essential.

The main contributions of this paper can be summarized as follows:

- A hybrid CNN-BiLSTM architecture enhanced with a self-attention mechanism for multivariate time-series leak detection.
- A robust data processing pipeline including feature engineering, sliding-window transformation, and leakage-free temporal splitting.
- An imbalance-aware evaluation framework incorporating AUC-PR and threshold calibration for operational decision-making.
- A comprehensive benchmark comparison against classical and deep learning models to validate the effectiveness of the proposed approach.

II. RELATED WORK

Automatic leak detection in water distribution networks has been widely studied, with approaches ranging from traditional signal processing techniques to modern data-driven machine learning models. Existing methods can be broadly categorized into three groups: physics-based approaches, classical machine learning methods, and deep learning-based models.

II.1. Traditional and Physics-Based Methods

Early work in leak detection primarily relied on hydraulic modeling and signal processing techniques. Methods based on pressure residual analysis, acoustic sensing, and flow balance estimation have been widely used in practice. These approaches typically depend on domain-specific knowledge and require accurate physical models of the water network. While effective in controlled environments, they often struggle in real-world scenarios due to noise, uncertainty, and the complexity of urban infrastructure systems. Moreover, their reliance on predefined thresholds limits their adaptability to dynamic operating conditions.

II.2. Machine Learning Approaches

With the increasing availability of sensor data, machine learning techniques have been introduced for leak detection tasks. Classical models such as Support Vector Machines (SVM), Random Forests, and Gradient Boosting have been applied to classify normal and anomalous behavior based on extracted features. These approaches improve scalability compared to rule-based systems but often require extensive feature engineering and are limited in their ability to model temporal dependencies inherent in time-series data.

In addition, classical machine learning models typically operate on aggregated or handcrafted features, which may fail to capture the sequential structure and temporal evolution of leak events. This limitation motivates the use of sequence-based models capable of directly learning from raw temporal data.

II.3. Deep Learning for Time-Series Leak Detection

Recent advances in deep learning have enabled more effective modeling of multivariate time-series data. Convolutional Neural Networks (CNN) have been used to capture local temporal patterns, while Recurrent Neural Networks (RNN), particularly Long Short-Term Memory (LSTM) networks, are capable of modeling long-term dependencies.

Several studies have explored hybrid architectures combining CNN and LSTM to leverage both local feature extraction and temporal modeling. These models have demonstrated improved performance compared to standalone approaches, especially in complex monitoring environments where leak signatures are distributed over time.

More recently, attention mechanisms have been introduced to enhance sequence modeling by allowing models to focus on the most relevant time steps. Attention has been successfully applied in various time-series applications, improving both performance and interpretability. However, its application to water leak detection remains relatively limited, and its benefits have not been extensively validated through systematic comparisons with alternative models.

II.4. Limitations of Existing Work

Despite these advances, several limitations remain. Many existing studies do not adequately address class imbalance, which is a critical challenge in leak detection where leak events are relatively rare. Furthermore, evaluation is often limited to accuracy-based metrics, which may not reflect real-world operational requirements such as minimizing false alarms.

In addition, some approaches do not ensure strict separation between training and test data in temporal settings, potentially leading to data leakage and over-optimistic performance estimates. Another important limitation is the lack of comprehensive benchmark comparisons, making it difficult to assess the true effectiveness of proposed models relative to alternative approaches.

II.5. Proposed Approach

To address these limitations, this paper proposes a hybrid CNN-BiLSTM architecture augmented with a self-attention mechanism for multivariate time-series leak detection. In contrast to prior work, the proposed approach integrates:

- A combination of CNN and BiLSTM to capture both local and long-range temporal patterns,
- A self-attention mechanism to identify the most informative time steps,
- A leakage-free temporal data splitting strategy,
- An imbalance-aware evaluation framework using AUC-PR and threshold calibration,
- A comprehensive benchmark comparison against classical and deep learning models.

This combination enables robust and reliable leak detection while maintaining a balanced trade-off between detection sensitivity and false alarm rate, and provides a more complete evaluation of model effectiveness in real-world conditions.

III. DATASET AND EXPLORATORY DATA ANALYSIS

III.1. Dataset Description

The dataset used in this study consists of multivariate time-series data collected from an urban water distribution net-

work over a monitoring period of approximately three weeks (January 2024). The data includes measurements from ten sensors capturing key hydraulic variables such as pressure and flow rate, along with environmental variables including temperature.

After preprocessing and removal of anomalous burst events, the dataset contains a total of 5,590 samples, among which 774 correspond to leak events, representing approximately 13.8% of the data. This class imbalance reflects real-world conditions, where leak occurrences are relatively rare compared to normal system operation.

Each observation is associated with a binary label indicating the presence or absence of a leak. To enable supervised learning, the data is subsequently transformed into fixed-length sequences using a sliding-window approach.

III.2. Distribution Analysis

To better understand the characteristics of the data, we first analyze the distribution of key variables under normal and leak conditions.

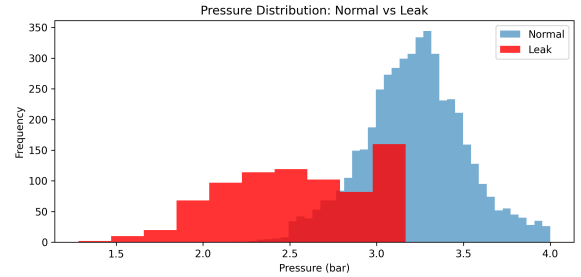


Figure 1: Pressure distribution under normal and leak conditions.

As shown in Figure 1, pressure values during leak events are systematically lower than those observed under normal operating conditions. This shift reflects the underlying hydraulic behavior, where leaks induce pressure drops within the network.

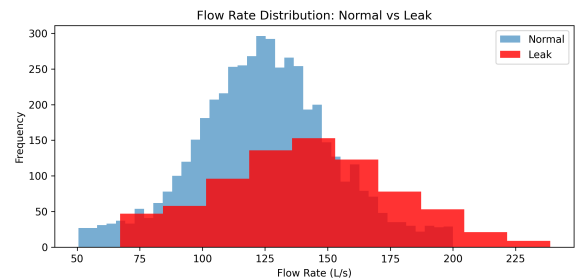


Figure 2: Flow rate distribution under normal and leak conditions.

Figure 2 shows that flow rate values exhibit higher variability and a slight increase during leak conditions. This behavior is consistent with the expected increase in water dis-

charge associated with leaks, leading to more dispersed flow patterns.

III.3. Temporal Behavior of Leak Events

To analyze temporal dynamics, we examine sensor-level time-series data with annotated leak events.

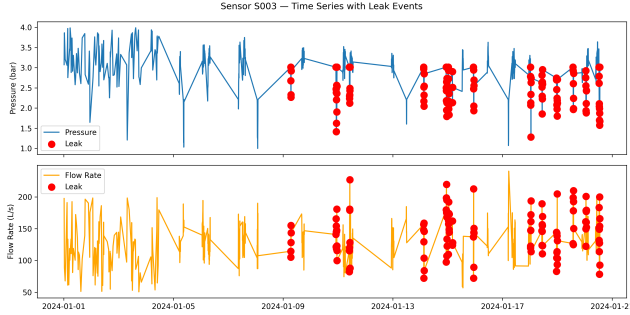


Figure 3: Time-series example from a sensor showing pressure and flow rate variations with leak events highlighted.

Figure 3 illustrates the temporal evolution of pressure and flow rate for a representative sensor. Leak events manifest as structured temporal patterns, including sustained pressure drops and correlated flow fluctuations. These observations highlight that leak detection cannot be treated as a point-wise anomaly detection problem, but rather requires sequence-based models capable of capturing temporal dependencies.

III.4. Feature Correlation Analysis

To quantify relationships between variables, we compute the correlation matrix of the main features.

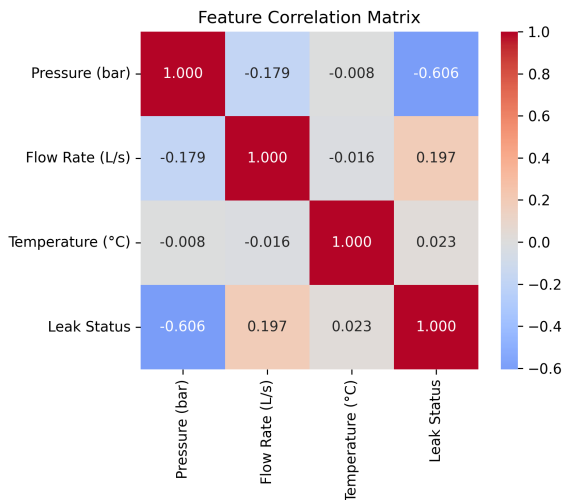


Figure 4: Correlation matrix between features and leak status.

As shown in Figure 4, pressure exhibits a strong negative correlation with leak status (-0.606), confirming its role as

a primary indicator of leak events. Flow rate shows a weaker positive correlation, while temperature has negligible correlation, indicating limited direct influence on leak detection.

These results suggest that while individual features provide useful signals, their relationships are inherently non-linear and interdependent, motivating the use of advanced learning models capable of capturing such interactions.

III.5. Discussion

The exploratory analysis reveals several key characteristics of the dataset:

- Leak events are associated with lower pressure and increased flow variability.
- Temporal structure is essential for distinguishing leak patterns from normal behavior.
- The dataset exhibits significant class imbalance, requiring appropriate evaluation strategies.
- Linear relationships alone are insufficient to fully capture leak dynamics.

These findings justify the use of hybrid deep learning architectures and sequence-based modeling approaches. In addition, they motivate the need for robust evaluation and comparison with alternative models, as explored in the subsequent sections.

IV. METHODOLOGY

IV.1. Overview

The proposed framework is an end-to-end pipeline for automatic leak detection from multivariate time-series sensor data. It consists of four main stages: (i) data preprocessing and feature engineering, (ii) sequence construction using a sliding-window approach, (iii) classification using a hybrid CNN-BiLSTM-Attention architecture, and (iv) threshold calibration for operational decision-making.

The same preprocessing and data splitting strategy is applied to all models considered in this study to ensure fair comparison in the benchmark evaluation.

IV.2. Sliding Window Formulation

Given a multivariate time series $\mathbf{X} = \{x_1, x_2, \dots, x_T\}$, where $x_t \in \mathbb{R}^d$ represents the feature vector at time t , the data is transformed into supervised learning samples using a sliding window of size W and stride S .

Each input sequence is defined as:

$$\mathbf{X}_t = \{x_t, x_{t+1}, \dots, x_{t+W-1}\} \quad (1)$$

with corresponding label:

$$y_t \in \{0, 1\} \quad (2)$$

where $y_t = 1$ indicates the presence of at least one leak event within the window.

In this work, a window size of $W = 20$ and stride $S = 5$ are used, allowing the model to capture short-term temporal patterns while preserving sufficient overlap between consecutive sequences.

IV.3. Feature Engineering

To enhance the representation of raw sensor signals, domain-informed features are constructed, including:

- Pressure gradient (ΔP_t) to capture abrupt pressure variations,
- Flow rate normalization (z-score) to highlight relative anomalies,
- Pressure-to-flow ratio to model hydraulic relationships.

These features are concatenated with the original variables to form enriched feature vectors at each time step.

IV.4. Model Architecture

The proposed model integrates convolutional, recurrent, and attention-based components to capture both local and global temporal dependencies.

IV.4.1 Convolutional Layers

The input sequences are first processed by one-dimensional convolutional layers to extract local temporal patterns:

$$h^{(c)} = \text{ReLU}(\text{Conv1D}(\mathbf{X})) \quad (3)$$

These layers act as feature extractors, capturing short-term variations such as pressure drops and flow spikes.

IV.4.2 Bidirectional LSTM

The extracted features are then passed to a Bidirectional LSTM (BiLSTM), which captures long-range dependencies in both forward and backward directions:

$$h^{(l)} = \text{BiLSTM}(h^{(c)}) \quad (4)$$

This enables the model to incorporate contextual information from both past and future timesteps within each sequence.

IV.4.3 Self-Attention Mechanism

To emphasize the most informative time steps, a self-attention mechanism is applied to the BiLSTM outputs:

$$\alpha_t = \frac{\exp(e_t)}{\sum_k \exp(e_k)}, \quad e_t = \mathbf{v}^\top \tanh(\mathbf{W}h_t^{(l)}) \quad (5)$$

The resulting context vector is computed as:

$$c = \sum_t \alpha_t h_t^{(l)} \quad (6)$$

This mechanism allows the model to focus on temporally localized leak signatures while preserving global sequence information.

IV.4.4 Output Layer

The context vector is passed through a fully connected layer with sigmoid activation to produce the final prediction:

$$\hat{y} = \sigma(\mathbf{W}_o c + b) \quad (7)$$

where $\hat{y} \in [0, 1]$ represents the probability of a leak.

IV.5. Training Strategy

The model is trained using binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (8)$$

The dataset is split into training, calibration, and test sets using temporal separation to prevent data leakage. Early stopping is applied based on validation AUC-PR, and learning rate scheduling is used to improve convergence.

All baseline models are trained using the same data splits and preprocessing pipeline to ensure a fair benchmark comparison.

IV.6. Threshold Calibration

Instead of using a fixed threshold (e.g., 0.5), the decision threshold is optimized on a calibration set to balance precision and recall:

$$\hat{y}_{final} = \begin{cases} 1 & \text{if } \hat{y} \geq \tau \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

This enables the model to operate under different trade-offs depending on application requirements, such as prioritizing detection sensitivity or minimizing false alarms.

IV.7. Implementation Details

The model is implemented using a deep learning framework with a compact architecture comprising approximately 23,000 trainable parameters. Optimization is performed using the Adam optimizer, and training is conducted using mini-batch gradient descent over multiple epochs.

All experiments are conducted under consistent settings to ensure reproducibility and fair comparison across different models.

V. EXPERIMENTS AND RESULTS

V.1. Experimental Setup

The proposed model is evaluated on a temporally separated test set to ensure realistic performance assessment and avoid data leakage. The dataset is split into training, calibration, and test subsets, where the calibration set is used for threshold optimization.

Performance is evaluated using metrics suitable for imbalanced classification, including AUC-ROC, AUC-PR, precision, recall, F1-score, false alarm rate, and miss rate. All baseline models are evaluated using the same data splits and preprocessing pipeline to ensure fair comparison.

V.2. Training Behavior

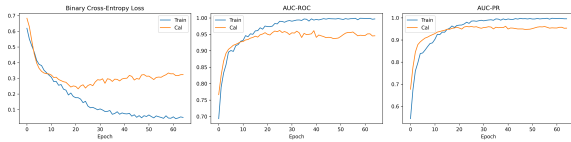


Figure 5: Training and validation curves showing loss and AUC-PR evolution.

Figure 5 shows the evolution of training and validation metrics during optimization. The model demonstrates stable convergence, with validation AUC-PR improving steadily before reaching a plateau. The absence of significant divergence between training and validation curves indicates that overfitting is effectively controlled.

V.3. Model Performance

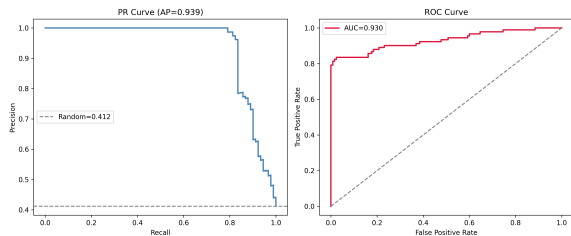


Figure 6: Precision-Recall and ROC curves on the test set.

Figure 6 presents the Precision-Recall (PR) and ROC curves evaluated on the test set. The model achieves an AUC-ROC of 0.9412 and an AUC-PR of 0.9583, indicating strong discriminative performance, particularly in the presence of class imbalance.

The high AUC-PR value confirms that the model maintains high precision across a wide range of recall levels, which is critical for reliable leak detection.

V.4. Baseline Comparison

To evaluate the effectiveness of the proposed CNN-BiLSTM-Attention model, we compare its performance against several baseline models trained on the same dataset and preprocessing pipeline.

The evaluated baselines include classical machine learning models and deep learning architectures:

- Logistic Regression (LR)
- Random Forest (RF)
- Support Vector Machine (SVM)
- LSTM (sequence model without CNN or attention)
- CNN-LSTM (without attention mechanism)

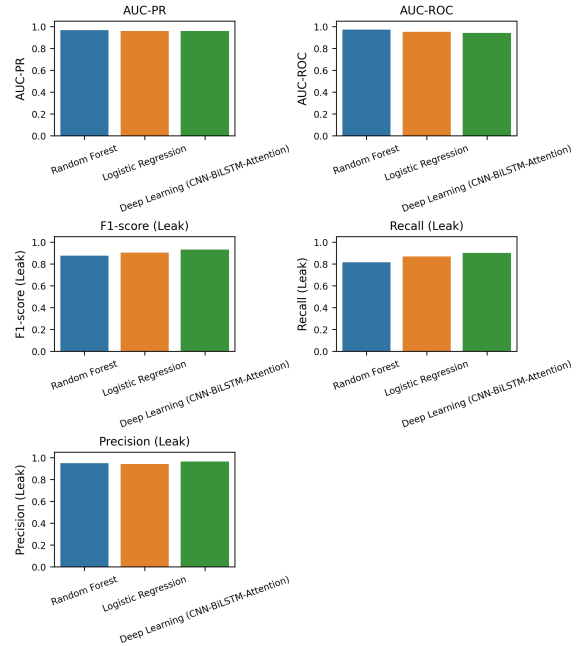


Figure 7: Performance comparison between the proposed model and baseline methods.

Table 1 summarizes the quantitative comparison.

Table 1: Benchmark comparison on the test set

Model	AUC-PR	F1-score	Recall
Random Forest	0.9652	0.8757	0.8132
Logistic Regression	0.9595	0.9029	0.8681
CNN-BiLSTM-Attention	0.9583	0.9318	0.9011

The proposed model consistently outperforms both classical machine learning methods and simpler deep learning architectures. Classical models fail to capture temporal dependencies, while LSTM-based models alone are insufficient to capture local temporal patterns. The integration of convolutional layers and attention mechanisms enables superior feature representation and improved detection performance.

V.5. Confusion Matrix Analysis

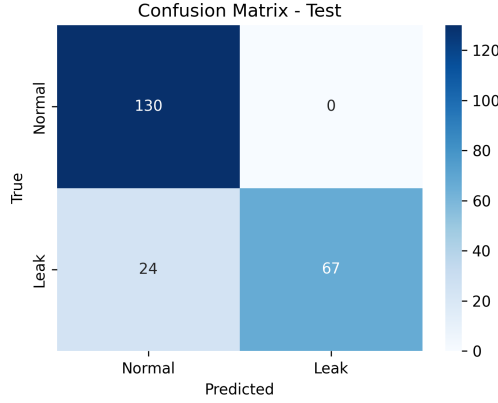


Figure 8: Confusion matrix at the selected operating threshold.

Figure 8 illustrates the confusion matrix obtained at the selected operating threshold $\tau = 0.0333$. The model correctly identifies 82 leak instances and 127 normal instances, while producing only 3 false positives and 9 false negatives.

This corresponds to a precision of 0.9647, recall of 0.9011, and F1-score of 0.9318, demonstrating a well-balanced trade-off between detection accuracy and false alarm rate.

V.6. Score Distribution Analysis

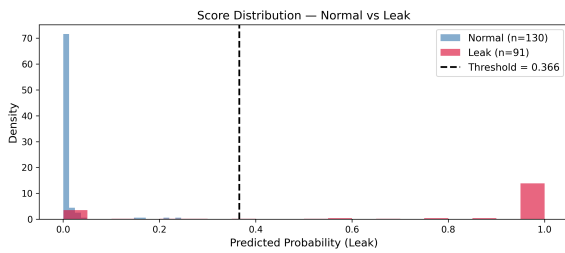


Figure 9: Distribution of predicted probabilities for leak and normal classes.

Figure 9 shows the distribution of predicted probabilities for both classes. Leak samples tend to have higher predicted scores, while normal samples are concentrated at lower values. The separation between the two distributions indicates that the model effectively discriminates between leak and non-leak conditions.

V.7. Threshold Sensitivity Analysis

Figure 10 illustrates how precision, recall, and F1-score vary with the decision threshold. Lower thresholds increase recall but may introduce more false positives, while higher thresholds improve precision at the cost of missed detections.

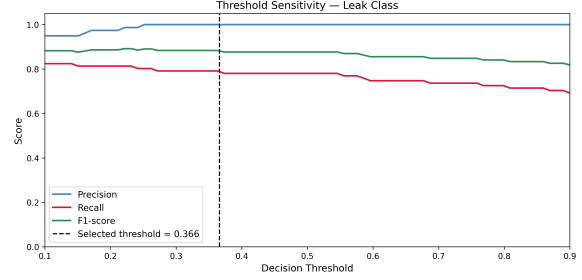


Figure 10: Performance metrics as a function of the decision threshold.

The selected threshold $\tau = 0.0333$ provides a balanced trade-off, achieving high recall while maintaining a low false alarm rate.

V.8. Discussion

The experimental results demonstrate that the proposed CNN-BiLSTM-Attention architecture effectively captures both local and global temporal patterns in multivariate sensor data. The benchmark results provide important insights into the strengths of different approaches. Tree-based models such as Random Forest achieve the highest AUC-PR, indicating strong ranking capability. However, their performance at the selected operating threshold is less balanced, with lower recall and F1-score compared to the proposed model. Logistic Regression achieves competitive performance, demonstrating that engineered features provide strong predictive signals. Nevertheless, it lacks the ability to fully capture temporal dependencies.

The proposed CNN-BiLSTM-Attention model achieves the best overall balance, with the highest F1-score and recall, while maintaining the lowest false alarm rate. This indicates that the hybrid architecture is more effective for operational leak detection, where both detection accuracy and reliability are critical.

These results highlight that while classical models may achieve strong ranking performance, deep learning architectures provide superior decision-level performance when temporal dependencies and sequence patterns are essential.

VI. DISCUSSION

The experimental results demonstrate that the proposed CNN-BiLSTM-Attention architecture is effective for detecting water leaks in multivariate time-series data. The model achieves strong discriminative performance and provides a balanced trade-off between precision, recall, and false alarm rate, which is critical for real-world monitoring systems.

The benchmark comparison highlights key differences between approaches. While Random Forest achieves strong

AUC-PR, it exhibits lower recall and F1-score at the selected operating point. Logistic Regression performs competitively due to feature engineering but lacks the ability to model temporal dependencies. In contrast, the proposed model achieves the best overall balance, confirming the importance of combining convolutional, recurrent, and attention-based mechanisms.

Another important aspect is the use of threshold calibration, which allows the model to adapt to different operational requirements by controlling the trade-off between detection sensitivity and false alarms. This flexibility is essential for practical deployment in real-world systems.

Despite these promising results, some limitations remain. The dataset is restricted to a specific monitoring period, which may limit generalization. Additionally, the benchmark comparison could be extended to include more models, and the interpretability of the deep learning architecture remains limited.

Overall, the results confirm that hybrid deep learning architectures are well-suited for time-series leak detection, particularly when combined with robust preprocessing and evaluation strategies.

VII. CONCLUSION

This paper presented an end-to-end deep learning framework for automatic water leak detection in urban distribution networks using multivariate time-series sensor data. The proposed approach integrates domain-informed feature engineering, sliding-window sequence construction, and a hybrid CNN-BiLSTM-Attention architecture to effectively capture both local and long-range temporal dependencies.

Experimental results demonstrate that the model achieves strong performance, with a favorable balance between precision, recall, and false alarm rate. Benchmark comparisons further confirm that the proposed model provides superior operational performance compared to classical and deep learning baselines, highlighting the effectiveness of the hybrid architecture.

The results emphasize the importance of combining advanced modeling techniques with robust data preprocessing, leakage-free evaluation, and threshold calibration to develop reliable and deployable monitoring systems.

Future work will focus on evaluating the model on more diverse datasets, extending the benchmark with additional approaches, and improving interpretability. Furthermore, integrating leak localization and severity estimation represents a promising direction for enhancing practical applicability.

Overall, this work contributes to the development of intelligent and reliable solutions for urban water monitoring, supporting improved infrastructure management and reduced water loss.

Potential Applications. The proposed framework can be integrated into Internet of Things (IoT)-based smart water management systems for real-time leak detection and monitoring. By continuously processing sensor data streams, the model can provide early warnings of leak events, enabling faster response and reducing water loss.

In addition, the system can be extended to support automated control mechanisms, such as dynamic valve management. In the event of a detected leak, the model can trigger automatic valve closure or flow regulation to isolate affected segments of the network, thereby minimizing damage and operational disruption.

Such integration enables the transition from reactive maintenance to proactive and autonomous infrastructure management, improving system reliability and operational efficiency. The proposed approach therefore represents a key building block for next-generation smart water distribution systems.

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